Chapter 12, Problem 36P

Problem

A 4200-V, three-phase transmission line has an impedance of 4 + *j* Ω per phase. If it supplies a load of 1 MVA at 0.75 power factor (lagging), find:

(a) the complex power

(b) the power loss in the line

(c) the voltage at the sending end

Step-by-step solution

Step 1 of 3

(A) $S = 1\left[0.75 + Sin(Cos^{-1}0.75)\right] = 0.75 + j0.6614 mVA$.

Step 2 of 3

(B) $\overline{S} = 3V_p I_p^* \Rightarrow I_p^* = \frac{S}{3V_p}$

$= \frac{(0.75 + j0.6614)\times 10^6}{3\times 4200}$

$= 59.52 + j52.49$

$P_L = |I_p|^2 R_L = (79.36)^2(4)$

$= 25.19\ KW$.

Step 3 of 3

(C) $V_S = V_L + I_p(4+j)$

$= 4.4381 - j0.21 KV$

$= 4.443\angle -2.709°\ KV$.